

NICHOLAS BORRELLO

AI & Software Engineer — Agentic Systems & LLM Applications

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SKILLS & CAPABILITIES

AI & AGENTS

LLM agents • Model Context Protocol (MCP) • Tool calling • Structured outputs • Agent orchestration • Model routing/fallback • RAG • LLM evaluation • VLM/OCR

LANGUAGES & ML

TypeScript • Python • Java • SQL • PyTorch • CNNs • Transformers • spaCy • MediaPipe • Computer vision • Document intelligence

FULL STACK & INFRASTRUCTURE

React • Next.js • Node.js • Bun • Hono • PostgreSQL • SQLite • Supabase • Docker • Playwright • GitHub Actions • Azure DevOps

EXPERIENCE

Bay State Pet & Garden — AI & Software Engineer

Taunton, MA (Remote) | Jul 2024 – Present

- ◆ Develop an AI Product Intelligence system that researches external product data, resolves exact product/variant identity, and produces evidence-backed catalog changes for human review.
- ◆ Engineer the agent execution layer with 25 bounded research tools, model routing and fallback, privacy controls, tool/model budgets, deadlines, and fail-closed validation.
- ◆ Maintain the retail CMS and AI evaluation pipeline, testing new extraction and agent behavior against verified datasets before staged rollout into product onboarding and publishing workflows.

Allegro MicroSystems — Process Engineering Intern

Manchester, NH | Jun 2023 – Aug 2023

- ◆ Developed automated test and deployment services using Python, Java, and Docker to standardize driver updates across engineering equipment.
- ◆ Maintained Azure DevOps CI/CD pipelines and Node.js automation utilities, replacing manual deployment processes.

EDUCATION

Worcester Polytechnic Institute — M.S. Computer Science

Worcester, MA | Aug 2025

Worcester Polytechnic Institute — B.S. Computer Science, *With Distinction*; Dean's List (Fall 2022)

Worcester, MA | May 2024

SELECTED PROJECTS

Resumancer AI — Agentic Career Workspace

Oct 2025 – Present | Next.js, TypeScript, Vercel AI SDK, Supabase

- ◆ Built a career workspace comparing job requirements with a Master CV, proposing evidence-backed changes with reviewable diffs instead of unconstrained rewrites.
- ◆ Added ATS compatibility checks, model fallback, and PDF export so verified career data moves from job analysis to an application-ready resume.

ShopSite MCP Server — E-Commerce Tool Integration

Jan 2026 | TypeScript, Model Context Protocol, Zod, Node.js

- ◆ Built an MCP server that lets AI agents retrieve orders, search products, and manage inventory in ShopSite without interacting directly with legacy APIs.
- ◆ Wrapped XML/CGI requests, HMAC-SHA1 authentication, and legacy response formats behind typed, validated tools designed for safe LLM use.

American Sign Language Gesture Recognition

Jan 2025 – May 2025 | Python, PyTorch, ResNet34, MediaPipe

- ◆ Trained CNN & Transformer models to recognize 100+ ASL gestures using ResNet34 and MediaPipe landmark sequences.
- ◆ Engineered temporal sequence extraction pipelines and parallelized data preprocessing using Python multiprocessing.

NEVI Search Tool — Policy Document Intelligence

Aug 2022 – Dec 2022 | Python, spaCy, PyMuPDF, PyQt6

- ◆ Built an NLP-assisted document search tool for locating policy evidence across lengthy state NEVI plans using spaCy and keyword/fuzzy matching.
- ◆ Implemented exact keyword search, fuzzy matching, configurable term groups, and spaCy preprocessing to speed review of lengthy state NEVI plans.